

COMPUTATIONAL FLUID DYNAMICS LAB

VII Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE51	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES:

Students undergoing this course are expected:

1. To study and analyze the internal and external aerodynamic properties over major aerodynamic profiles using commercial software packages.
2. To develop a Matlab code to analyze the behavior of the governing equations of Aerodynamics, Thermodynamics, Heat transfer, Aerospace Propulsion etc., and to generate major types of grid over an aerodynamic profile.

LIST OF EXPERIMENTS

- 1 Modelling of basic geometries using sketcher in workbench
- 2 Flow over an airfoil
- 3 2D analysis of road vehicles
- 4 2D analysis of high rise building
- 5 Supersonic flow over a wedge
- 6 Supersonic flow over missiles
- 7 Solution for one dimensional wave equation using explicit method of lax (code development)
- 8 Solution for one dimensional transient heat conduction equation using explicit method (code development)
- 9 Generation of the algebraic grids (code development)
- 10 Generation of the elliptic grids (code development)

Reference Books:

1. ANSYS FLUENT Tutorial Guide 18.0
2. Stormy Attaway, 'MATLAB-A Practical Approach'

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

- 1 Generate geometries or models to solve the problems using computational tools
- 2 Analyze flow around the various geometries such as airfoils, building and vehicles.
- 3 Develop MATLAB codes to solve physical problems governed by pdes.
- 4 Develop MATLAB codes to generate algebraic grids and elliptic grids over airfoil.
- 5 Evaluate the supersonic flow over 2d bodies.